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(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU, RTMNU & MSBTE Mumbai
Department of Computer Science & Engineering
"A place to learn; A chance to grow"
2025-2026(Odd Semester)

MSE-II EXAMINATION

Course: B. Tech in CSE — **Year/Semester:** 6^{th} Semester (3^{rd} Year)
Subject: Digital Image Processing — **Subject Code:** CS5T003
Max Marks: 20 — **Date: 01/04/2026** — **Duration: - 1 Hr.**

Instructions to the Students:
1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:
CO 1: understand the fundamental concepts of a digital image processing system.
CO 2: Analyse images in the frequency domain using various transforms
CO 3: Evaluate the techniques for image enhancement and image restoration.
CO 4: Categorize various compression techniques.
CO 5: Interpret Image compression standards.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	State the need for image compression	L1/CO4	1
Q.2	Enlist all arithmetic operation between images. + − × ÷	L1/CO3	1
Q.3	Extend JPEG and MPEG	L2/CO4	1
Q.4	Define Negative Transformation in Gray level transformation.	L1/CO3	1

SECTION B

1. **All Questions are compulsory.**
2. **Solve answer in brief (two, three Sentence).**

Q.5 Solve the Addition operation in arithmetic operator between images. — L3/CO3 — 2

Q.6 Design the histogram Equalizer for 8*8 image — L5/CO3 — 2

Gray Levels	0	1	2	3	4	5	6	7
No. of Pixels	80	100	90	60	30	20	10	0

Q.7 Solve by using Run length encoding for compressing Binary string 000000111111111111111000000000000011111111. — L3/CO4 — 2

Q.8 Compare Lossless and Lossy Compression Technique. — L4/CO4 — 2

SECTION C

1. **Solve any two questions.**

Q.9 Identify the Huffman code for the Probability 0.30, 0.25, 0.20, 0.12, 0.08, 0.05 — L1/CO4 — 4

Q.10 Solve and Perform the histogram equalization for an 8*8 image shown below — L3/CO3 — 4

Gray Levels	0	1	2	3	4	5	6	7
No. of Pixels	9	8	11	4	10	15	4	3

Q.11 Explain the logarithmic (log) transformation in Gray-level transformation with a suitable example with perfect C. — L5/CO3 — 4